

High-Efficiency Multiphase Modified DC-DC Cuk Converter

A. R. Assidiq
School of Electrical Engineering and Informatics
Bandung Institute of Technology
Bandung, Indonesia
anamraisassidiq@gmail.com

J. Furqani
School of Electrical Engineering and Informatics
Bandung Institute of Technology
Bandung, Indonesia
jfurqani@itb.ac.id

A. Rizqian
School of Electrical Engineering and Informatics
Bandung Institute of Technology
Bandung, Indonesia
windra@stei.itb.ac.id

Abstract—The problem in DC-DC power converters penetrating new and renewable energy is the ripple of high current, non-continuous current, and low efficiency. This paper proposes a multiphase modification DC-DC Cuk converter to obtain a continuous current on the input and output sides with predetermined component values and duty cycles. This power converter proposal aims to get a DC-DC power converter with a continuous input and outputs current, small current ripples, and high efficiency by looking at a fraction of power losses in the converter. The current ripple in this proposed converter is close to zero using a duty cycle value of 0.5, so there is ripple cancellation in the L_m inductor. The largest loss fraction lies in the switch, so efficiency improvements are obtained by replacing the IGBT switch with a MOSFET looking at better specifications. So that an efficiency increase was obtained in the proposed converter by 90.81% in the B circuit and 90.07% in the C circuit. Verification has been performed from the experiment.

Keywords—DC-DC Converter; Cuk; Modifications; Multiphase; Efficiency.

I. INTRODUCTION

The Indonesian government has set a target for the New and Renewable Energy (NRE) mix of 23% by 2025 and 31% by 2030 through the National Energy Policy and the National Energy General Plan. Meanwhile, in 2022, the NRE mix in Indonesia has only reached 12.70%. This can be attributed to Indonesia's lack of maximum in utilizing the renewable energy mix. One form of renewable energy that has great potential to be developed in Indonesia is solar power. National Energy General Plan even explained that one of the efforts to accelerate NRE in Indonesia is prioritized through solar power plants with its initiation in East Nusa Tenggara as a solar energy granary [1]. According to the Ministry of Energy and Mineral Resources of Indonesia, it was explained that the potential for solar power in Indonesia is 207.8 GW. Still, it was only installed at 0.2 GW at the end of 2021 [2].

Maximizing the use of solar power in achieving the energy mix in Indonesia is very important to pay attention to. The implementation of the use of solar power can use a microgrid system. That way, the built microgrid system becomes more environmentally friendly and fuel-efficient [3]. The microgrid system cannot be separated from the needs of the DC-DC converter in the system. In general, the power converters widely used today are composed of the base cells of both Buck and Boost DC-DC power converters. However, the power converter has the disadvantage of an input current that is not continuous, an output voltage with large ripples, requires but on the input and output side, and a large power loss on the filter. So, the DC-DC power converter is needed that can overcome these weaknesses.

This research will focus on implementing the modified Cuk converter with multiphase conditions previously studied [4]. The Cuk converter has a low output on the input and output sides because the current generated is continuous on both sides, even though the polarity of the output voltage is reversed compared to the input voltage. At the same time, its modification and multiphase techniques will influence the output's ripples, efficiency, and polarity. Then this study will also compare the topology of the multiphase Cuk converter that has been modified to obtain a higher efficiency value with current ripples close to zero. In the paper [5], it was found that the addition of a duty cycle would reduce efficiency, so in this study only alpha 0.5 was used to obtain a small current ripple.

II. PROPOSED CONVERTER

A. DC-DC Cuk Converter

The Cuk converter is a power converter that combines two converters, namely boost and buck converters, or a combination of the two converters. It is shown in Fig. 1. as the original Cuk converter.

The characteristic of this Cuk converter is that it has a continuous input and output current so that the ripples owned are low, but the polarity of the output voltage is reversed. In the circuit, a very high voltage gain is obtained at a higher output voltage [4]. So it is necessary to make another modification to the converter of Fig. 1. as shown in Fig. 2.

The result of the modifications in the series of Fig. 2. can provide the result of smaller input and output current ripples, the same voltage polarity as the input, as well as greater voltage gain for the same duty cycle value [4]-[6].

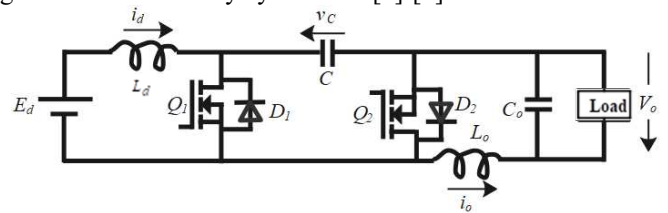


Fig. 1. DC-DC Cuk Converter [4]

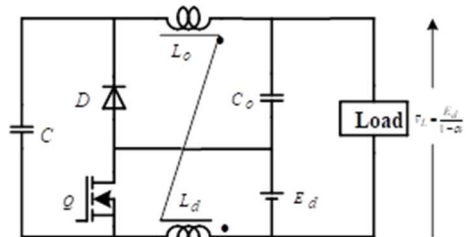


Fig. 2. DC-DC Converter Cuk Modification [4]

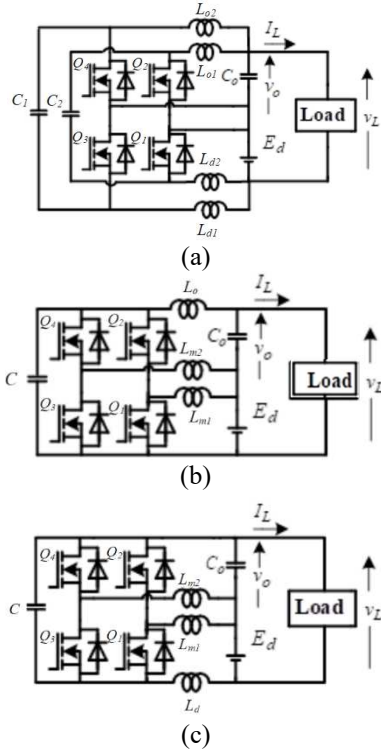


Fig. 3. DC-DC Converter Cuk Multiphase Modification [4] (a) Circuit A; (b) Circuit B; (c) Circuit C

B. Multiphase Modified Cuk DC-DC Converter

One way to reduce current ripples in a power converter is to use the multiphase technique [7]-[11]. The multiphase technique can be done on the modified Cuk converter circuit in Fig. 2. so that the following circuit will be obtained as shown in Fig. 3.

In Fig. 3. there are three pieces of multiphase modification Cuk series for 2 phases, which can then be further developed into an N-phase. It can be seen that the number of components used in a multiphase circuit will be 2 times compared to the previous circuit as in Fig. 3(a)., and the number of transistors will be 2 times that of a multiphase circuit as in Fig. 3(b). and Fig. 3(c). The use of a multiphase is intended to get the smallest possible current ripple at the output. So that the expected efficiency also increases. There are also differences in the placement of inductors in each circuit as shown in Fig. 3. Naming the inductor according to its position, o means "ON" or above, m means in the "MIDDLE", d means "DOWN" or below.

The output voltage for each of the circuits in Fig. 3. can be written as follows [4]:

$$V_L = \frac{E_d}{1-\alpha} - \frac{V_Q\alpha + V_D(1-\alpha)}{1-\alpha} - \frac{NR_d(1-2\alpha+2\alpha^2) + R_Q\alpha + R_D(1-\alpha)}{N(1-\alpha)^2} I_L \quad (1)$$

$$V_L = \frac{E_d}{1-\alpha} - \frac{V_Q\alpha + V_D(1-\alpha)}{1-\alpha} - \frac{R_m + R_oN(1-\alpha)^2 + R_Q\alpha + R_D(1-\alpha)}{N(1-\alpha)^2} I_L \quad (2)$$

$$V_L = \frac{E_d}{1-\alpha} - \frac{V_Q\alpha + V_D(1-\alpha)}{1-\alpha} - \frac{NR_m\alpha^2 + R_d + R_Q\alpha + R_D(1-\alpha)}{N(1-\alpha)^2} I_L \quad (3)$$

From equations (1), (2), and (3) it can be observed that the voltage drop value in inductors and transistors can be reduced by adding the number of N-phases. V_L is the output voltage at load, E_d is the input voltage, α is the duty cycle value, V_D V_Q is the voltage on the diode and transistor, and R_d R_D R_Q is the resistance in the inductor, diode, and transistor. In equation (2) it can be seen that the circuit Fig. 3(b). gives a smaller voltage drop value than the circuit of Fig. 3(c). This is because the value of the current flowing in the inductor L_o in the circuit of Fig. 3(b). has a smaller value than the value of the current flowing in the inductor L_d in the circuit of Fig. 3(c). When compared to the Circuit of Fig. 3(a). then the voltage shrinkage remains greater in the circuit of Fig. 3(c).

C. Voltage Gain Analysis

The voltage gain on the proposed converter can be searched using the state space equation. In this study, state space 2 was used with the condition that the transistors Q1 and Q4 will be ON together and the transistors Q2 and Q3 will be OFF together with a phase difference of 180°. The voltage gain decrease through the circuit Fig. 3(b). as follows.

State space 1: Q1 and Q4 OFF, Q2 and Q3 ON, for $0 < t < \alpha T_s$

$$\frac{d}{dt} \begin{bmatrix} i_{d1} \\ i_{d2} \\ i_o \\ v_c \\ v_o \end{bmatrix} = \begin{bmatrix} -\frac{R_d + R_Q}{L_d} & 0 & 0 & 0 & 0 \\ 0 & -\frac{R_d + R_D}{L_d} & 0 & -\frac{1}{L_d} & 0 \\ 0 & 0 & -\frac{R_o}{L_o} & \frac{1}{L_o} & -\frac{1}{L_o} \\ 0 & \frac{1}{C} & -\frac{1}{C} & 0 & 0 \\ 0 & 0 & \frac{1}{C_o} & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{d1} \\ i_{d2} \\ i_o \\ v_c \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{E_d - V_Q}{L_d} \\ \frac{E_d - V_D}{L_d} \\ \frac{E_d}{L_o} \\ 0 \\ -\frac{I_L}{C_o} \end{bmatrix} \quad (4)$$

State space 2: Q1 and Q4 ON, Q2 and Q3 OFF, for $\alpha T_s < t < 1$

$$\frac{d}{dt} \begin{bmatrix} i_{d1} \\ i_{d2} \\ i_o \\ v_c \\ v_o \end{bmatrix} = \begin{bmatrix} -\frac{R_d + R_D}{L_d} & 0 & 0 & -\frac{1}{L_d} & 0 \\ 0 & -\frac{R_d + R_Q}{L_d} & 0 & 0 & 0 \\ 0 & 0 & -\frac{R_o}{L_o} & \frac{1}{L_o} & -\frac{1}{L_o} \\ \frac{1}{C} & 0 & -\frac{1}{C} & 0 & 0 \\ 0 & 0 & \frac{1}{C_o} & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{d1} \\ i_{d2} \\ i_o \\ v_c \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{E_d - V_D}{L_d} \\ \frac{E_d - V_Q}{L_d} \\ \frac{E_d}{L_o} \\ 0 \\ -\frac{I_L}{C_o} \end{bmatrix} \quad (5)$$

If averaging is done in equations (4) and (5) equations (6) will be obtained.

$$0 = \begin{bmatrix} -\frac{R_d + R_Q\alpha + R_D(1-\alpha)}{L_d} & 0 & 0 & -\frac{1-\alpha}{L_d} & 0 \\ 0 & -\frac{R_d + R_Q\alpha + R_D(1-\alpha)}{L_d} & 0 & -\frac{\alpha}{L_d} & 0 \\ 0 & 0 & -\frac{R_o}{L_o} & \frac{1}{L_o} & -\frac{1}{L_o} \\ \frac{1-\alpha}{C} & \frac{\alpha}{C} & -\frac{1}{C} & 0 & 0 \\ 0 & 0 & \frac{1}{C_o} & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{d1} \\ i_{d2} \\ i_o \\ v_c \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{E_d - V_Q\alpha - V_D(1-\alpha)}{L_d} \\ \frac{E_d - V_D\alpha - V_Q(1-\alpha)}{L_d} \\ \frac{E_d}{L_o} \\ 0 \\ -\frac{I_L}{C_o} \end{bmatrix} \quad (6)$$

From equation (6) an inverse matrix will be carried out so that the output voltage value V_o is obtained as follows.

$$V_o = \frac{E_d (1 - \alpha)}{1 - \alpha} - \frac{V_Q \alpha + V_D (1 - \alpha)}{1 - \alpha} - \frac{R_d + R_o 2(1 - \alpha)^2 + R_Q \alpha + R_D (1 - \alpha)}{2(1 - \alpha)^2} I_L \quad (7)$$

From equation (7) it will be used to calculate the voltage at the load as follows.

$$V_L = E_d + V_o$$

$$V_L = \frac{E_d}{1 - \alpha} - \frac{V_Q \alpha + V_D (1 - \alpha)}{1 - \alpha} - \frac{R_d + R_o 2(1 - \alpha)^2 + R_Q \alpha + R_D (1 - \alpha)}{2(1 - \alpha)^2} I_L \quad (8)$$

R_d R_D R_Q R_o is the resistance in the inductor, diode, transistor, and output. It can be seen that in equation (8), there are 3 components of voltage gain, namely the original voltage gain, internal drop voltage, and parasitic components. Because the study used a duty cycle value of 0.5, equation $(1 - \alpha = \alpha)$ was obtained as follows.

$$V_L = \frac{E_d}{\alpha} - \frac{V_Q \alpha + V_D \alpha}{\alpha} - \frac{R_d + R_o 2\alpha^2 + R_Q \alpha + R_D \alpha}{2\alpha^2} I_L \quad (9)$$

D. Ripple Cancellation

To be able to find out the power losses in the power converter due to parasitic elements in each component as in equation (6), it is necessary to know the value of the current flowing in the component and obtain the DC value of each inductor as follows.

$$I_{d1} = I_{d2} = \frac{I_L}{2\alpha} \quad (10)$$

$$I_o = I_L \quad (11)$$

The representation of the total current on each of the components of the inductor is as follows:

$$I_{d1} = I_{d2} = I_{d1} = I_{d1DC} + I_{d1AC} \quad (12)$$

$$I_{d1} = \frac{I_L}{2\alpha} + \frac{1}{2\alpha L_d} [(R_d + R_Q \alpha + R_D \alpha) I_L + 2\alpha (E_d - V_Q \alpha - V_D \alpha - V_C \alpha)] \Delta t \quad (13)$$

$$\Delta t = \alpha T_s = \frac{\alpha}{f_s} \quad (14)$$

I_{d1} I_{d2} I_L I_o is the current in inductors 1, 2, and output. It is found that the value of the representation of the current in each inductor in the equation (13). From the inductor current on L_{d1} and L_{d2} , there will be ripple cancellation for a duty cycle value of 0.5. It is seen in Fig. 5. that there is a ripple cancellation. Ripple cancellation occurs as a result of the ripple reduction of each inductor current L_{d1} and L_{d2} as shown in Fig. 5. In addition to the duty cycle value of 0.5, it will result in an increase in current ripple in the inductor.

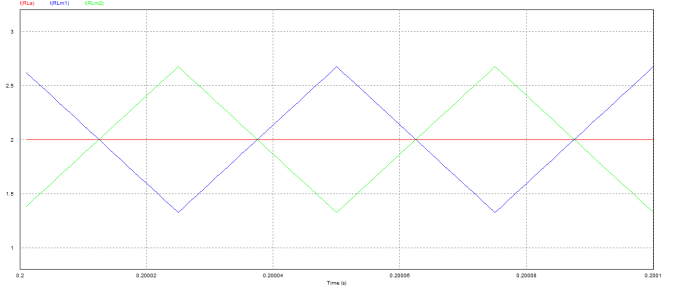


Fig. 5. Ripple Cancellation on Current Inductor

E. Power Loss Analysis

For power losses in the inductor sourced from the winding on the inductor can be modeled as follows.

$$P_L = \bar{I}^2 \cdot R_{L,DC} + \frac{\Delta i^2}{12} R_{L,AC} \quad (15)$$

$R_{L,DC}$ $R_{L,AC}$ is the resistance in DC and AC component. In equation (15) there are 2 components of power losses. The first part represents losses due to DC resistance and the second part represents losses due to AC resistance.

In this study, the core of the Toroid inductor KS200-060A was used with the core material being Sendust. The equation of power losses in the core of the inductor on the datasheet can be seen in equation (16).

$$P_{core} = \left(\frac{\Delta B}{2000} \right)^{2.225} \cdot (4,584 f_{SW} + 0,0238 f_{SW}^{1.996}) I_{Lo} \quad (16)$$

ΔB is the magnetic field change in the inductor, f_{SW} is the frequency switching. The conduction power losses on the switch are sourced from the presence of DC resistance on the switch and the internal voltage drop when the switch is in an ON or conduction condition. Modeling of conduction power losses on the switch can be made as follows.

$$P_Q = (i_{RMS}^2 \cdot R_Q + V_Q \cdot \bar{I}) \cdot \alpha \quad (17)$$

$$P_D = (i_{RMS}^2 \cdot R_D + V_D \cdot \bar{I}) \cdot (1 - \alpha) \quad (18)$$

Modeling of power losses due to the declaration process can be written as follows.

$$P_{SW} = (E_{ON} \cdot i_{Q,RMS} + E_{OFF} \cdot i_{Q,RMS}) \cdot f_{SW} \quad (19)$$

In equation (19) 2 components describe the power losses in the declaration process. The E_{ON} section describes the power loss that occurs when the switch changes from off to ON. The E_{OFF} section describes the power loss that occurs when the switch changes from an ON to an OFF state. Both parts are function values of the RMS current flowing on the switch so it is necessary to do a curve fitting from the characteristics of the switch used.

III. RESULT AND ANALYSIS

In the circuit, there is a gate driver as isolation between the converter prototype circuit and the declared microcontroller circuit. The microcontroller used in this experiment is the TMS320F28377D which can provide a box signal as input for the gate driver.

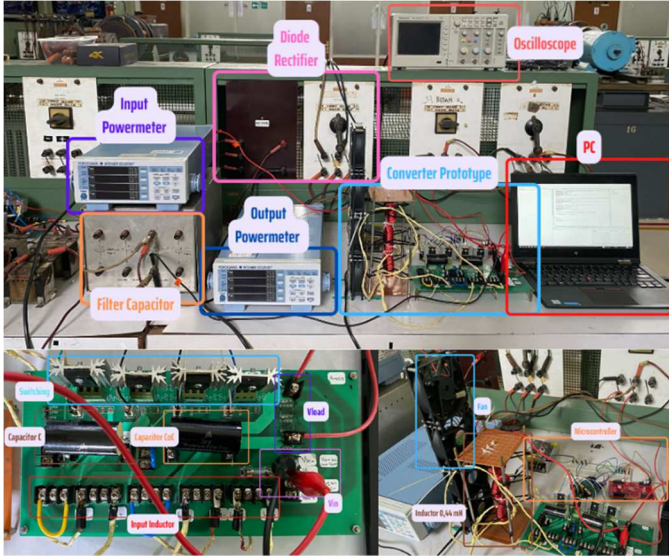


Fig. 6. Converter Prototype Testing Scheme

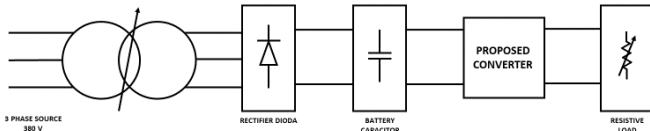


Fig. 7. Scheme of the Converter Prototype Circuit

There are two capacitors of the same type, namely the JFX Film Cap capacitor of 12 μF . Then three inductors with the type KS200-060A type inductor core are used. The inductance magnitude of the inductor used in this experiment was the same at 0.44 mH. The amount of DC and AC resistance in the inductor measured using the Sanwa LCR700 at a frequency of 10 kHz obtained an RDC value of 0.113 Ω and a RAC of 0.114 Ω . The classing frequency used in this experiment was made at 20 kHz with load variations from 2 A to 4 A. Scheme test series of prototype converters as shown in Fig. 6. and 7. are as follows.

The Cuk converter prototype of this study is a scale down from previous studies for the series of Fig. 3(b) and 3(c) because it has a very small current ripple value compared to the circuit of Fig. 3(a) [4]. The operating parameters can be seen as shown in Table 1.

TABLE I. OPERATION PARAMETERS

Component	Value/Model
Capasitors C and C _o	12 μF
Frequency Switching	20 kHz
Inductors L _m and L _o	0,44 mH
Switch	IGBT KGF50N60KDA
	MOSFET IRF260N
Power (Rating)	90 W – 200 W (150 W)
Maximum Ripple Current	1,3 A

In the experiment of the prototype converter series, various switches were also carried out to provide optimal results on efficiency. 2 switches were used, namely the first switch with the IGBT FGH40N60UFD type [12] and the second switch with the MOSFET IRF260N type [13]. The difference between both switches can be seen as shown in Table 2.

TABLE II. SWITCH COMPARISON

Parameter	IGBT FGH40N60UFD	MOSFET IRF260N
Switching Time	Medium	High
Diode Parasitic	Resistance in RC-IGBT	Resistance in Diode
Thermal Impedance	Relatively Low	Low, can provide better power dissipation
Rise dan Fall Time	Relatively Low	Low
Frequency	Low, below 20 kHz	High, up to above 200 kHz
Voltage	High, above 1 kV	Low, below 250 V
Market Price	Cheap	Relatively Cheap

A. Verification of Ripple Cancellation

As in Fig. 8. and Fig. 9. it is seen that the shape of the current ripples in the inductors L_{d1} and L_{d2} will be mutually canceled so that the ripple value of the L_o inductor current resembles the ripple value of the output current close to zero. The experiment is taken on a scale of 1 A/div and 100 mV/div. This is following the simulation in Fig. 5. This is because, for the duty cycle value of 0.5, a current ripple value of 1.3 A will be obtained, each of which values on the L m inductor reduce each other so that a zero value is obtained. Whereas the inductor L_o has a ripple value of zero current.

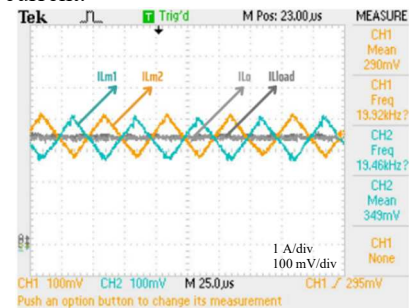


Fig. 8. Circuit Inductor Current Wave 3(b).

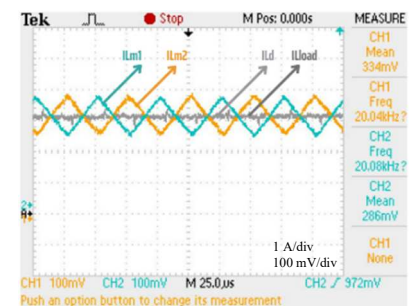


Fig. 9. Circuit Inductor Current Wave 3(c).

B. Fraction Loss Power Converter

Loss of power on the prototype converter proposed has the largest fraction of power loss on the switch which can see in Fig. 10. and Fig. 11. This is because the duty cycle value of 0.5 gives a small current ripple to the output current so that the effect of power losses on the inductor is small. And the impact is on the switch with the biggest losses being in the transistors and diodes in the switch. So that by changing the switch used in the experiment, it will provide an improvement in the efficiency of the prototype converter. The first switch will be used, namely IGBT, then the repair uses the second switch, namely MOSFET. In testing this power fraction, it still uses the first switch, namely IGBT.

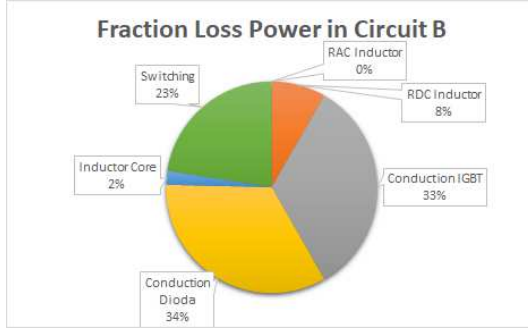


Fig. 10. Circuit Power Loss Fraction of Circuit Fig. 3(b).

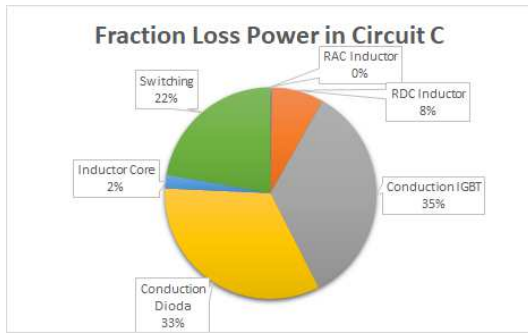


Fig. 11. Circuit Power Loss Fraction of Circuit Fig. 3(c).

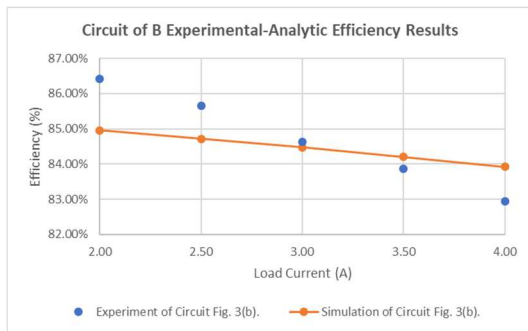


Fig. 12. Efficiency of Prototype Circuit Fig. 3(b).

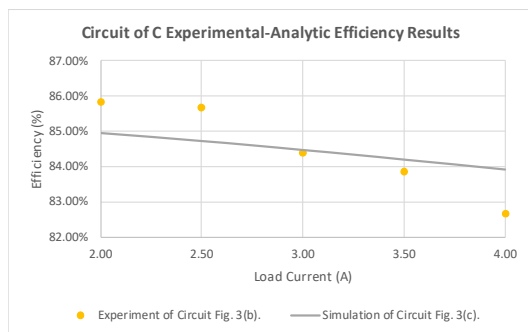


Fig. 13. Efficiency of Prototype Circuit Fig. 3(c).

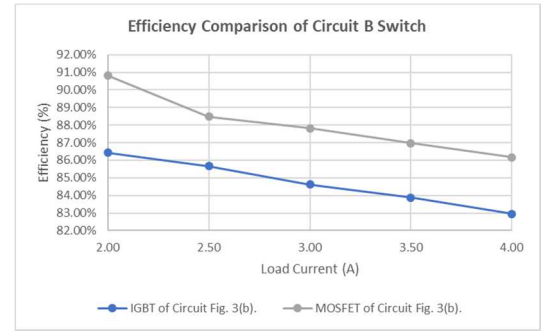


Fig. 14. Efficiency Improvement After Circuit Fig. 3(b). Switch Replacement

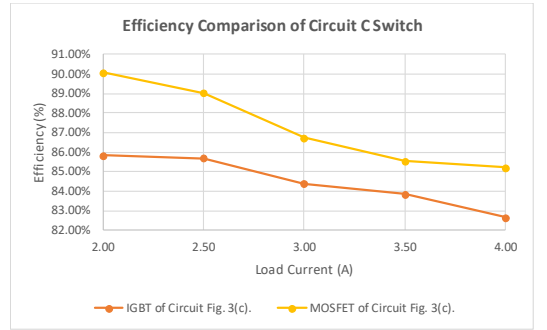


Fig. 15. Efficiency Improvements After Circuit Fig. 3(c). Switch Replacement

C. The Efficiency of the Proposed Converter

The results of the efficiency of the calculation analytics and experiments. This test found that the trend of decreasing efficiency in the calculation results with the experiment results was the same; when there was an increase in load, the efficiency of the converter would decrease slowly. This is because the addition of the load modeled with the increase in load current will undoubtedly cause a quadratic increase in power losses due to the parasitic resistance of each component. The highest efficiency is obtained when the current load value is still low, namely when the input voltage is 24 V and the current load value is 2 A, with the maximum efficiency obtained being 86.43% for Circuit Fig. 3(b). and 85.82% for the Circuit Fig. 3(c).

After efficiency improvements were made, namely by changing the IGBT switch to a MOSFET switch, an efficiency increase of 4% was obtained as shown in Fig. 13. It was found that changing the IGBT switch to a MOSFET switch will provide an increase in converter efficiency as shown in Fig. 14. and Fig. 15. This is because the specification of the MOSFET has better components in terms of DC resistance on transistors and diodes, and also has a lower internal fall voltage compared to IGBT. Then another reason is that the MOSFET is indeed suitable for use at low voltages with a high declaring frequency. At the same time, IGBT is suitable at high voltages with a low declaring frequency below 20 kHz. So with MOSFET, it can provide its advantages in the prototype series of converters.

IV. CONCLUSION

The multiphase-modified DC-DC Cuk converter with 2 phases can be used as a DC-DC power converter instead of other conventional DC-DC converters such as Boost, Buck, or Cuk converters. This is because the multiphase-modified DC-DC Cuk converter can provide current and voltage on the continuous input and output side, at a duty cycle value of 0.5, it can provide zero current ripples due to ripple cancellation, has the same output voltage polarity as the input voltage, and has high efficiency with before the repair in the B circuit which is 86.43% and the C circuit which is 85.82% and after

improvement in the B circuit to 90.81% and the C circuit to 90.07% where can in Fig. 14. and Fig. 15. The biggest losses on the destination converter lie in the switch so the replacement of the switch will increase the efficiency of the speed converter.

ACKNOWLEDGEMENT

This first author would like to thank Institute Technology of Bandung (ITB) as the primary funder of this research.

REFERENCE

- [1] Direktorat Jenderal Energi Baru Terbarukan dan Konservasi Energi (EBTKE), "Forum Kehumasan Dewan Energi Nasional: Menuju Bauran Energi Nasional Tahun 2025," 2022, in press.
- [2] Berita Universitas Pertamina, "Potensi Energi Surya Indonesia Capai 207,8 Giga Watt, Kapasitas Terpasang Baru 200,1 Mega Watt," 2022, in press.
- [3] N. Hatzigiargyriou, H. Asano, R. Iravani and C. Marnay, "Microgrids," in *IEEE Power and Energy Magazine*, vol. 5, no. 4, pp. 78-94, July-Aug. 2007.
- [4] A. Dahono, "Konverter Cuk Yang Dimodifikasi Sebagai Sel Dasar Berbagai Konverter Daya, ITB," 2021.
- [5] M. N. A. Rabbani, "Topologi Baru Konverter DC-DC Dua-Arah Dengan Riak Masukan Dan Keluaran Sangat Kecil, ITB," 2021.
- [6] A. Dahono, A. Rizqiawan, & P.A. Dahono. (2020). A modified Cuk DC-DC converter for DC microgrid systems. *TELKOMNIKA*, 18(6), 3247-3257.
- [7] C. Pan and C. Lai, "A High-Efficiency High Step-Up Converter with Low Switch Voltage Stress for Fuel-Cell System Applications," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 6, pp. 1998-2006, June 2010.
- [8] Qian L, Liang c, Jian W, Rui M, Dongdong Z, Fei G "An Improved Floating Interleaved *Boost* Converter With the Zero-Ripple Input Current for Fue Cell Applications," *IEEE Trans On Energy Conversion*, Vol. 34, No. 4, December 2019.
- [9] Joseph, A. Daniel, and A. Unnikrishnan, "Interleaved Cuk converter with improved transient performance and reduced current ripple," *The Journal of Engineering*, 2017, Vol. 2017, Iss. 7, pp. 362–369.
- [10] Y. Gu and D. Zhang, "Interleaved *Boost* converter with ripple cancellation," *IEEE Trans. Power Electron.*, 2013, 28, (8), pp. 3860–3869.
- [11] K. D. Joseph, A. E. Daniel, and A. Unnikrishnan, "Interleaved Cuk Converter with reduced switch current," *Int. Conf. on Power, Instrumentation Control and Computing*, 2018, pp. 1-6.
- [12] All Data Sheet, "FGH40N60UFD Datasheet (PDF) – Fairchild Semiconductor," unpublished.
- [13] All Data Sheet, "IRFP260N Datasheet (PDF) – International Rectifier," unpublished.